

Speech-Recognition

Exercises Pattern Recognition, Speech Signal

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Abstract

Physical properties of sounds

Practical Part 01 (preparation of midterm 01)

For this exercise a tool like Audacity is useful. If you haven't installed a sound editing tool on your computer, select and install one. It must be possible to edit the sound, insert silence and change the format.

Record an own short speech command word (e.g. "yes", "no", "go", "up", "left", ...) to wave! Make it as similar as possible with the wav-files from the speech_command dataset by cutting out some parts (milliseconds) or inserting silence. Make at least 5 recordings.

Theory

Part 01: Nyquist frequency and rate

It is said that birds may sing with a frequency up to 8 kHz.

What is the Nyquist frequency and the Nyquist rate for sampling such a bird song.

Part 02: Data of digitalized sounds

- a) Given the results of part 01, calculate the amount of data generated for a 1 minute recording of a birds song. The sound shall be recorded mono (1 channel) with 1 Byte per sample.
- b) If MP3 can achieve a compression rate of 75% and 30 minutes are saved in a MP3 file. How big is the file?

Part 03: MFCC

Visit the website <https://haythamfayek.com/2016/04/21/speech-processing-for-machine-learning.html> 
(visited 24/04/05)

and work through the description. Especially, find out the x- and y- axes of the diagrams:

- Waveform
- Spectrum / Mel-scale spectrum
- Spectrogram
- MFCC diagram

If you want, you may test and update the used Python code.